RS05 Motor Notes for Bistable Hook Excitation

This document holds the RS05 motor details that were removed from the main bistable derivation. The derivation only keeps the actuator velocity-limit equations needed by the optimization model.

Motor Parameters

Parameter	Value	Notes
Rated torque	1.6 N·m	Continuous rating with heat sinking
Peak torque	5.5 N·m	Short-duration command limit
No-load output speed	480 RPM ±10%	$\omega_{\max} \approx 50.3$ rad/s at the output shaft
Torque constant K_t	$0.94\ \mathrm{N} \cdot \frac{\mathrm{m}}{A_{\mathrm{rms}}}$	Converts Iq current to torque
Gear ratio	7.75:1	Planetary reduction
Drive mode	FOC	Field-oriented control
Maximum phase current	$11 A_{\mathrm{pk}}$	Torque is still limited to the peak torque range
CAN bus rate	1 Mbps	CAN 2.0B extended frame
Encoder	14-bit absolute	Used for snap detection and state feedback

Table 1: RS05 parameters used by the bistable simulations.

Velocity Limit

The 480 RPM value is a shaft angular velocity limit, not a direct frequency limit. For an oscillating shaft command at frequency f and angular amplitude Θ,

$$v_{\mathrm{peak}} = \Theta \cdot 2\pi f \leq \omega_{\max} \tag{1}$$

Therefore,

$$\Theta_{\max}(f) = \frac{\omega_{\max}}{2\pi f} \tag{2}$$

With a lever arm r, the maximum linear stroke is

$$\delta_{\max}(f) = \frac{\omega_{\max} r}{2\pi f} \tag{3}$$

CAN Control Modes

The run mode is selected by writing `run_mode` at index `0x7005` with Communication Type 18, then enabling the motor with Type 3.

Value	Mode	Use
0	Operation	MIT-style torque/position/velocity command. Set $K_p = K_d = 0$ for pure torque feed-forward through t_{ff}.
3	Current	Direct Iq command through `iq_ref`; no position or speed loop.
5	Position CSP	Cyclic synchronous position; useful only if the position trajectory and velocity limit are acceptable.

Table 2: RS05 control modes relevant to sinusoidal excitation.

Streaming A Sine Command

The motor firmware does not provide an onboard sine-wave generator. A controller must compute each sample and stream it over CAN.

For operation mode, send Type 1 frames with

$$t_{\mathrm{ff}}(t) = \tau_0 \sin(2\pi f t), \quad K_p = 0, \quad K_d = 0 \tag{4}$$

For current mode, write

$$i_{q(t)} = \frac{\tau_0}{K_t} \sin(2\pi f t) \tag{5}$$

to `iq_ref` at index `0x7006`. For example, a 2.87 N·m torque amplitude corresponds to about 3.05 A using $K_t = 0.94$ N·m/A.

ESP32 Timing

At 1 Mbps CAN, a 29-bit extended frame is roughly 110 bits, or about 110 μs on the bus. Two motors commanded at 500 Hz each require about 1000 frames/s, so the command traffic is roughly 11% bus utilization before feedback frames. A 1 ms hardware-timer loop on the ESP32 is adequate if WiFi and Bluetooth are disabled and the CAN/timer work is kept off latency-sensitive tasks.